

EARTHQUAKES

The study of earthquakes allows geologists to “guess” what is occurring below the Earth’s surface.

A **seismograph** is used to measure the vibrations caused by an earthquake.

These vibrations spread out from the **focus** of the quake - the point inside the Earth where the energy is first released that causes the earthquake.

The **epicentre** of the quake is the point on the Earth’s surface directly above the focus.

The rolling of the Earth that occurs during an earthquake is due to three different types of waves that travel out from the focus.

1. **COMPRESSION WAVES (P waves)**
 - First to reach the surface of the Earth.
 - Matter is squeezed together at the head of the wave, pulled apart behind it.
 - Move through solids, liquids and gases.
2. **SHEAR WAVES (S waves)**
 - Slower waves.
 - Vibrate side to side in an S-shaped motion.
 - **Cannot** pass through liquids.
3. **SURFACE WAVES (L waves)**
 - Slowest moving.
 - Travel along the surface of the Earth.
 - Result from the whole area vibrating .

The epicentre of a quake is determined by measuring the time difference between the arrival of the P and S waves.

This is done by three or more recording stations and the exact location is determined by the intersection of circles drawn on a map, each one centred on a station.

LOCATION OF EARTHQUAKES

Earthquakes tend to occur along faults when one part of the crust moves suddenly in relation to another part.

Massive amounts of energy are released during this shift and can cause major destruction and loss of life if the epicentre is near a major population centre.

If the earthquake occurs under the sea, **tsunamis** (wrongly called tidal waves) can occur, causing massive destruction to coastal regions and low-lying islands.